The Ephemeris

December 2020
Volume 31 Number 04- The Official Publication of the San Jose Astronomical Association

December 2020 - March 2021 Events
Schedule subject to change due to Covid-19 restrictions.
Please check our website for up to date information.

Board & Speaker Talk (online)
Saturday 12/5, 1/2, 1/30 2/27, 3/27
Board Meetings: 6 -7:30pm
Speaker Talk: 8-9:30pm

Fix-It Day (2-4pm)
Currently on hold due to COVID-19 Shelter in Place

Solar Sunday (1:30-3:30pm)
Sunday 12/2, 1/10, 2/7, 3/7

Intro to the Night Sky Class
Friday 12/14, 1/22, 2/19, 3/19

Houge Park 1Q and 3Q In-Town Star Party
Currently on hold due to COVID-19 Shelter in Place

RCDO Starry Nights Star Party
Currently on hold due to COVID-19 Shelter in Place

Imaging SIG Mtg
Tuesday 12/18, 1/19, 2/16, 3/16

Astro Imaging Clinics
(at Little Uvas OSP)
Currently on hold due to COVID-19 Shelter in Place

Binocular Star Gazing
Will resume in Summer 2021

Armchair Star Party
Dates TBA

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlight-ed in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible can-cellation due to weather.

INSIDE THIS ISSUE

Gecko Nebula
Photo by SJAA Member: **Francesco Meschia**
September 19, 2020
Location: Pinnacles National Park, Soledad, CA

Houge Park Facility Storage Cleanup

Muditha Kanchana—Kanch

Houge Park Facility Storage space is used by different sub programs within the SJAA community. Main users of the storage space are the Loaner, Solar, Donations, Quick Start, Imaging, and Library programs as well as the storage of administrative materials. We long had a need to organize the storage space. A proposal was presented by me to the Board & approved in mid 2019. Calls for adding more locked storage & racks was part of the proposal.

Few members Glenn, Akshay, Nikola & myself met up in the Houge Park building with full Corvid-19 precautions on Oct 10th, 2020. We spent a few hours cleaning, rearranging storage, and making use of the new locked cabinets. We now have most of our expensive or easy to misplace items in locked cabinets. Each program got designated cabinet & rack space.

Above L to R: Akshay, Nikola, Kanch and Glenn on October 10, 2020 at Houge Park showing the clean and organized room

November Armchair Star Party

Muditha Kanchana—Kanch

SJAA Armchair Star Party returned in November after a three-month hiatus from the summer series. We selected five objects to showcase this month; Mars, Wizard nebula, Elephant trunk nebula, Owl/ET cluster, and Andromeda galaxy. November hosts were Rashi, Carl, Nancy, Swami, Glenn, Marianne, Wolf, Nikola, Joe, Kanch as well as interns Chelsea & Sophie. We hope to return next year with more Armchair events.

Above: The armchair star party hosts for the November Armchair Star Party

SJAA wishes its members Happy Holidays and Best Wishes for a Happy New Year 2021!

Lemonade from lemons in the heart of Auriga

By Francesco Meschia

On the weekend of the October New Moon I drove down to Pinnacles West, defying the forecasts that envisioned still some residual smoke in the Salinas valley. I was joined there by two other imagers from SJAA.

I was carrying high hopes for my imaging session. For the last ten days or so I had been tweaking, cleaning and adjusting my equatorial mount to ensure tracking was as smooth as it could be, and I was very happy as I had managed to tune my RMS error to less than half an arcsecond. So I was looking forward to using, for the first time, my new TSFLAT25 flattener-with-no-reducer and enjoy imaging at a scale of 0.89 arcsec/pixel.

Driving south on 101 I could see the layer of smoke, apparently floating over Gilroy. The Salinas valley also seemed filled with hazy smoke, but it looked like it was mostly sitting in the lower layers of the atmosphere. With some luck, the 1900-ft elevation of the West side observing site would’ve put me out of the worst of it.

Setting up my rig by the visitors’ center, just before sunset, the transparency didn’t seem bad. As the twilight progressed, and the stars started coming out of the sky, my sky radiance meter was reading higher and higher figures, up to 21.2 - 21.3 magnitudes per square arcsecond by the end of the astronomical twilight – my plan had worked! Elated, I slewed the telescope to my primary target for the night: IC342, the “hidden galaxy”. Being quite a faint object, I had planned to image it for the entire night, 8 to 9 hours.

That, unfortunately, did not happen. While the transparency wasn't too bad, the seeing turned out to be much more "twinkly" than the forecast had advertised, and I was looking, sadly, at a circle of confusion due to the seeing much bigger than the half arcsecond I had hoped and tuned my mount for. The guiding graph was all over the place. I wasn’t too keen on wasting 8 hours on a target that might not be a good test for the new flattener after all, so I switched to my secondary target, the Bubble nebula, the same one I had tested at home. Being higher in the sky, the seeing seemed to be better. So, I decided to image it, at 0.89 arcsec/pixel, until it reached the meridian. Then, I would’ve examined the subs and decided whether to persist in the effort after a meridian flip.

After blinking through the subs I wasn't a very happy camper. There was good nebula data, the stars were round, but they were fat! I decided that it wasn't worth continuing and that, instead, the mediocre seeing was the universe telling me that I should image at a larger image scale for the night. So, I swapped my AT130EDT for the little AT65EDQ, and since Auriga had cleared the trees to the East by a good margin, I set out to collect as much data as I could from a region in that constellation that I had wanted to image for a long time.

The AT65EDQ had just the perfect field of view to capture both IC405 and IC410 together, with the bright and colorful stars 16, 17, 18, 19, and IQ Aurigae in between. It's a spot in the sky that I love: dramatic nebulae and even more dramatic contrast of colors. The five hours of data I collected was clean and easy to process, and I'm quite happy with the end result. I guess I made some lemonade out of the lemons that the seeing conditions had thrown my way.

Technical Details:

Telescope: AT65EDQ flat-field quadruplet refractor, 65mm aperture f/6.5

Mount: iOptron iEQ45 Pro

Camera: Nikon D5500Ha

Guider: Lacerta MGEN

37 x 480s light frames at ISO200, calibrated with bias, dark and flat frames

Software: Processed with PixInsight and Starnet++

The Sky Shed

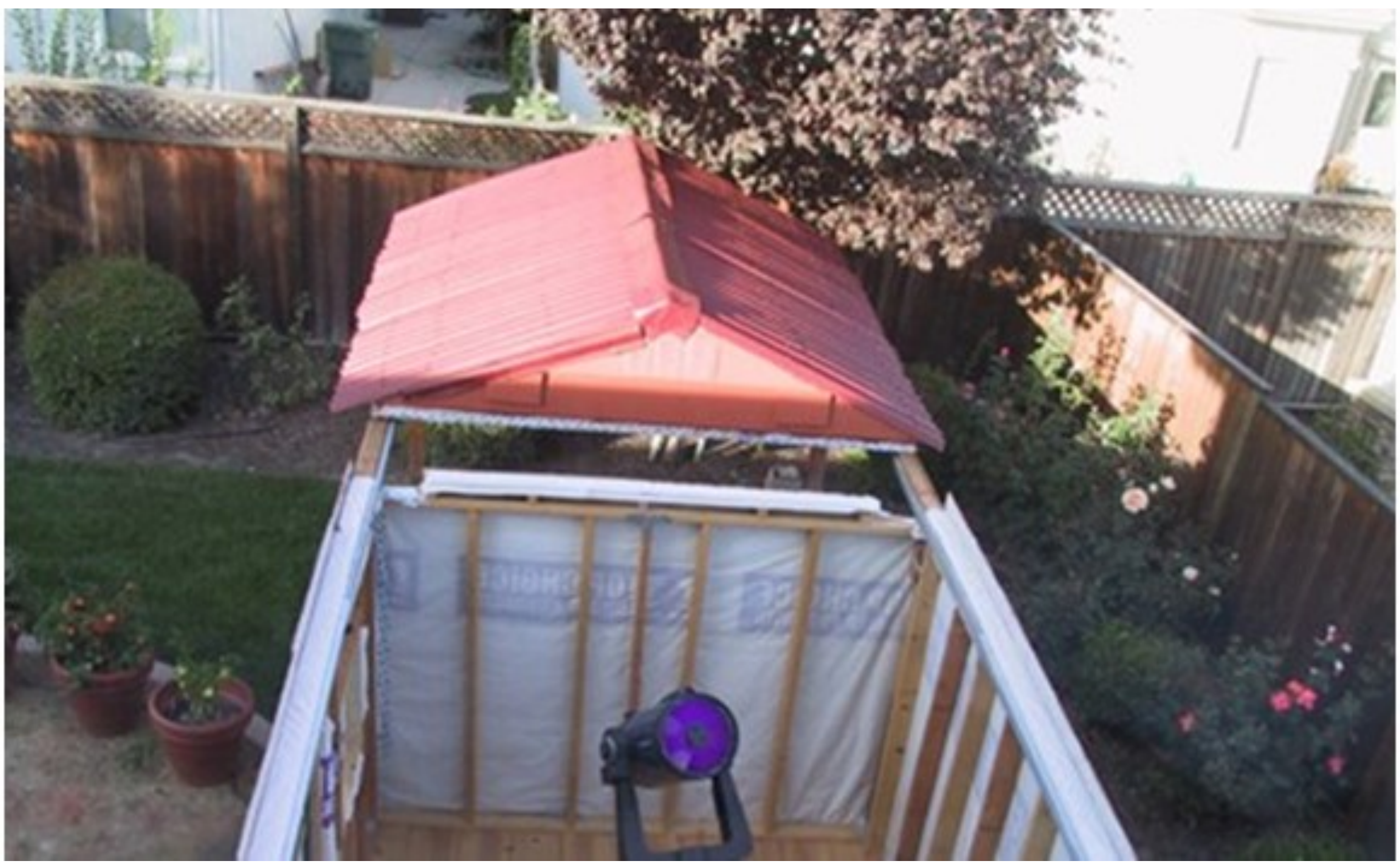
By Paul Kohlmiller (reprint from March 2009)

In 2004 I built a Sky Shed in my backyard. Not being possessed of the carpentry skills of Norm from This Old House, it was not a sure thing that I could do it. Just determining where to put it within my small backyard was not a sure thing. Most of the southern sky was going to be blocked by our house. Finally I decided that a particular location would give me a sliver of the southern sky so that I could see Sagittarius in June. The northern sky is blocked by a tree but those objects will spin overhead if I wait long enough.

The plans for the Sky Shed come from the Canadian company of the same name. Initially that was all they did, sell plans. If you're lucky you can find an affiliate of this company who will help build it for you but I'm not aware of anyone in the Bay Area who will do this. So I made many trips to Home Depot and Lowe's. I also had to find a garage door company that would sell me spare tracks. This is how the Sky Shed works, the roof rolls on garage door wheels and tracks.

Nearly all of the work on the shed was done on weekends and I didn't really get serious until May. I was pretty much finished by the end of August. A lot of the work shows creativity and ingenuity. Scratch that. A lot of the work shows improvisation based on the materials that I could find. For example, the Sky Shed plans assume that you can find a lumber company that will sell you 10" wide planks that are literally 10" wide. Very little lumber is sold that way, measure a 2x4 the next chance you get if you really think it is 2 inches by 4 inches. Also, Sky Shed recommends a steel roof but I thought that would make a lot of racket on rainy days and the shed is barely 6 feet from the family room window. I found some roof material made from tar which absolved me of the need to use tar paper on the roof. Rather than use wood planks for the exterior walls I used something called Hardie Panel which is a cement -based siding. I even made a door out of this material which was very unclever. But it does work.

The shed has worked well for more than 4 years. I think the folks at Sky Shed don't expect these sheds to last more than 10 years. You can find out more about the Sky Shed at http://www.skyshed.com.



A personal observatory is not a big deal. Some wood, some time, a lot of sweat, hammer-damaged thumbs, disbelieving spouse, curious neighbors and a damaged windshield from trying to put a 10 foot board into your 9 foot long van will get you at least halfway there. The result looks like it was done by an amateur but, hey, that's what we are. For better inspiration see www.skyshed.com. *Photo courtesy of Paul Kohlmiller.*

Earth and Moon Once Shared a Magnetic Shield, Protecting Their Atmospheres

Credit: NASA

Four-and-a-half billion years ago, Earth's surface was a menacing, hot mess. Long before the emergence of life, temperatures were scorching, and the air was toxic. Plus, as a mere toddler, the Sun bombarded our planet with violent outbursts of radiation called flares and coronal mass ejections. Streams of charged particles called the solar wind threatened our atmosphere. Our planet was, in short, uninhabitable.

The Earth and Moon, shown here in a composite of two images from the Galileo mission of the 1990s, have a long shared history. Billions of years ago, they had connected magnetic fields.

Credits: NASA/JPL/ USGS

But a neighboring shield may have helped our planet retain its atmosphere and eventually go on to develop life and habitable conditions. That shield was the Moon, says a NASA-led study in the journal Science Advances.

"The Moon seems to have presented a substantial protective barrier against the solar wind for the Earth, which was critical to Earth's ability to maintain its atmosphere during this time," said Jim Green, NASA's chief scientist and lead author of the new study. "We look forward to following up on these findings when NASA sends astronauts to the Moon through the Artemis program, which will return critical samples of the lunar South Pole."

A brief history of the Moon

The Moon formed 4.5 billion years ago when a Mars-sized object called Theia slammed into the proto-Earth when our planet was less than 100 million years old, according to leading theories. Debris from the collision coalesced into the Moon, while other remnants reincorporated themselves into the Earth. Because of gravity, the presence of the Moon stabilized the Earth's spin axis. At that time, our planet was spinning much faster, with one day lasting only 5 hours.

And in the early days, the Moon was a lot closer, too. As the Moon's gravity pulls on our oceans, the water is slightly heated, and that energy gets dissipated. This results in the Moon moving away from Earth at a rate of 1.5 inches per year, or about the width of two adjacent dimes. Over time, that really adds up. By 4 billion years ago, the Moon was three times closer to Earth than it is today – about 80,000 miles away, compared to the current 238,000 miles. At some point, the Moon also became "tidally locked," meaning Earth sees only one side of it.

Scientists once thought that the Moon never had a long-lasting global magnetic field because it has such a small core. A magnetic field causes electrical charges to move along invisible lines, which bow down toward the Moon at the poles. Scientists have long known about Earth's magnetic field, which creates the beautifully colored aurorae in the Arctic and Antarctic regions. The movement of liquid iron and nickel deep inside the Earth, still flowing because of the heat left over from Earth's formation, generates the magnetic fields that make up a protective bubble surrounding Earth, the magnetosphere.

But thanks to studies of samples of the lunar surface from the Apollo missions, scientists figured out that the Moon once had a magnetosphere, too. Evidence continues to mount from samples that were sealed for decades and recently analyzed with modern technology.

Like Earth, the heat from the Moon's formation would have kept iron flowing deep inside, although not for nearly as long because of its size.

"It's like baking a cake: You take it out of the oven, and it's still cooling off," Green said. "The bigger the mass, the longer it takes to cool off."

A magnetic shield

The new study simulates how the magnetic fields of the Earth and Moon behaved about 4 billion years ago. Scientists created a computer model to look at the behavior of the magnetic fields at two positions in their respective orbits.

At certain times, the Moon's magnetosphere would have served as a barrier to the harsh solar radiation raining down on the Earth-Moon system, scientists write. That's because, according to the model, the magnetospheres of the Moon and Earth would have been magnetically connected in the polar regions of each object. Importantly for the evolution of Earth, the high-energy solar wind particles could not completely penetrate the coupled magnetic field and strip away the atmosphere.

But there was some atmospheric exchange, too. The extreme ultraviolet light from the Sun would have stripped electrons from neutral particles in Earth's uppermost atmosphere, making those particles charged and enabling them to travel to the Moon along the lunar magnetic field lines. This may have contributed to the Moon maintaining a thin atmosphere at that time, too. The discovery of nitrogen in lunar rock samples support the idea that Earth's atmosphere, which is dominated by nitrogen, contributed to the Moon's ancient atmosphere and its crust.

Scientists calculate that this shared magnetic field situation, with Earth and Moon's magnetospheres joined, could have persisted from 4.1 to 3.5 billion years ago.

Earth and Moon… continued from Page 5

“Understanding the history of the Moon's magnetic field helps us understand not only possible early atmospheres, but how the lunar interior evolved,” said David Draper, NASA’s deputy chief scientist and study co-author. “It tells us about what the Moon's core could have been like -- probably a combination of both liquid and solid metal at some point in its history -- and that is a very important piece of the puzzle for how the Moon works on the inside.”

Over time, as the Moon’s interior cooled, our nearest neighbor lost its magnetosphere, and eventually its atmosphere. The field must have diminished significantly 3.2 billion years ago, and vanished by about 1.5 billion years ago. Without a magnetic field, the solar wind stripped the atmosphere away. This is also why Mars lost its atmosphere: Solar radiation stripped it away.

If our Moon played a role in shielding our planet from harmful radiation during a critical early time, then in a similar way, there may be other moons around terrestrial exoplanets in the galaxy that help preserve atmospheres for their host planets, and even contribute to habitable conditions, scientists say. This would be of interest to the field of astrobiology – the study of the origins of life and the search for life beyond Earth.

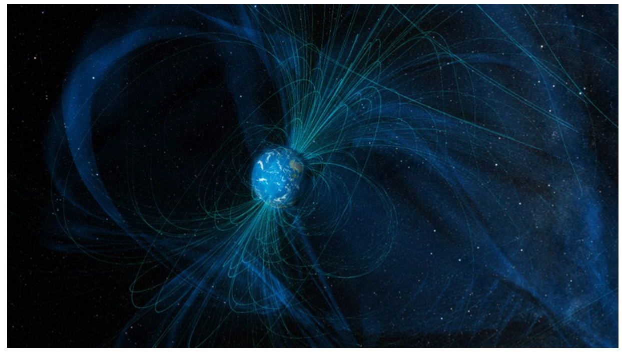
Human exploration can tell us more

This modeling study presents ideas for how the ancient histories of Earth and Moon contributed to the preservation of Earth’s early atmosphere. The mysterious and complex processes are difficult to figure out, but new samples from the lunar surface will provide clues to the mysteries.

As NASA plans to establish a sustainable human presence on the Moon through the Artemis program, there may be multiple opportunities to test out these ideas. When astronauts return the first samples from the lunar South Pole, where the magnetic fields of the Earth and Moon connected most strongly, scientists can look for chemical signatures of Earth’s ancient atmosphere, as well as the volatile substances like water that were delivered by impacting meteors and asteroids. Scientists are especially interested in areas of the lunar South Pole that have not seen any sunlight at all in billions of years -- the “permanently shadowed regions” – because the harsh solar particles would not have stripped away volatiles.

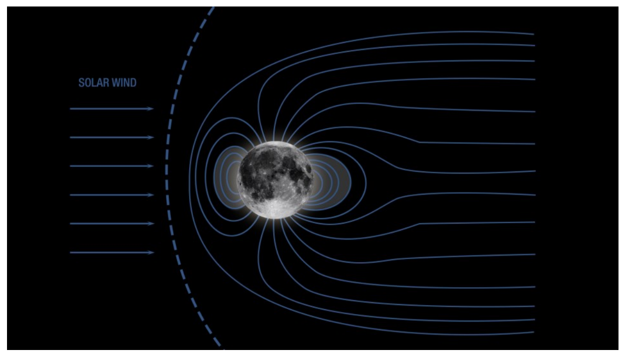
Nitrogen and oxygen, for example, may have traveled from Earth to Moon along the magnetic field lines and gotten trapped in those rocks.

“Significant samples from these permanently shadowed regions will be critical for us to be able to untangle this early evolution of the Earth’s volatiles, testing our model assumptions,” Green said.



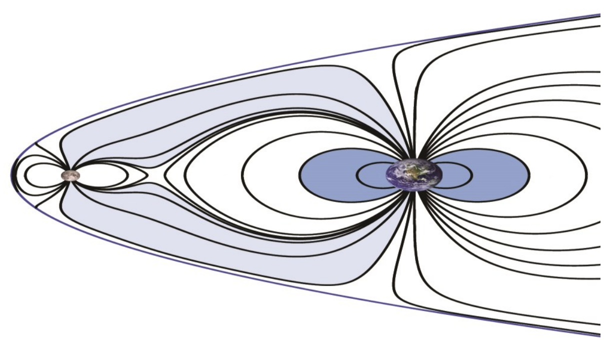
This illustration shows magnetic field lines that Earth generates today. The Moon no longer has a magnetic field.

Credits: NASA



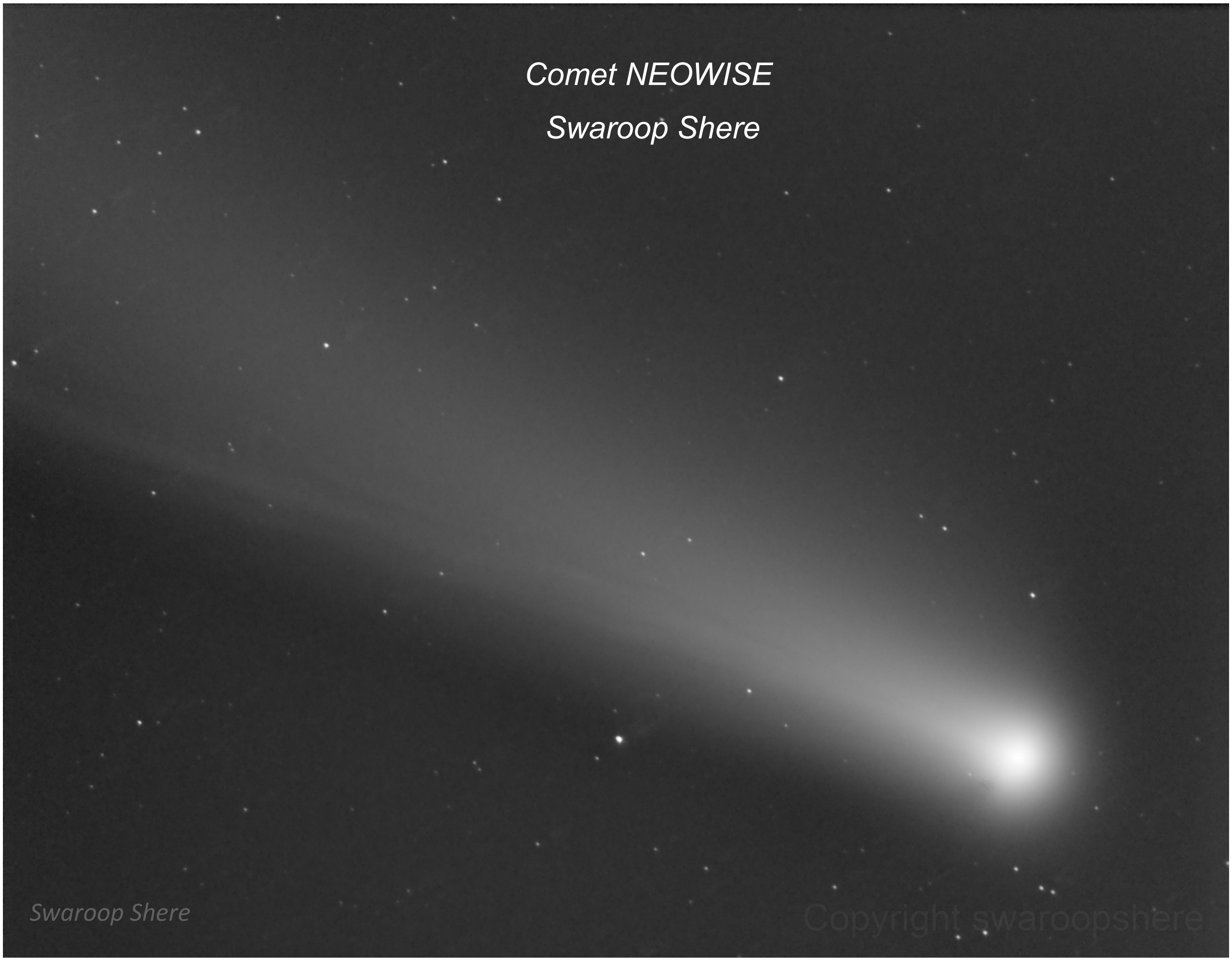
When the Moon had a magnetic field, it would have been shielded from incoming solar wind, as shown in this illustration.

Credits: NASA



This illustration shows how Earth and its Moon both had magnetic fields that were connected billions of years ago, helping to protect their atmospheres from streams of damaging solar particles, according to new research

Credits: NASA



Date: July 24, 2020

Location: Stellar Skies Observatory, Pontotoc, Tx

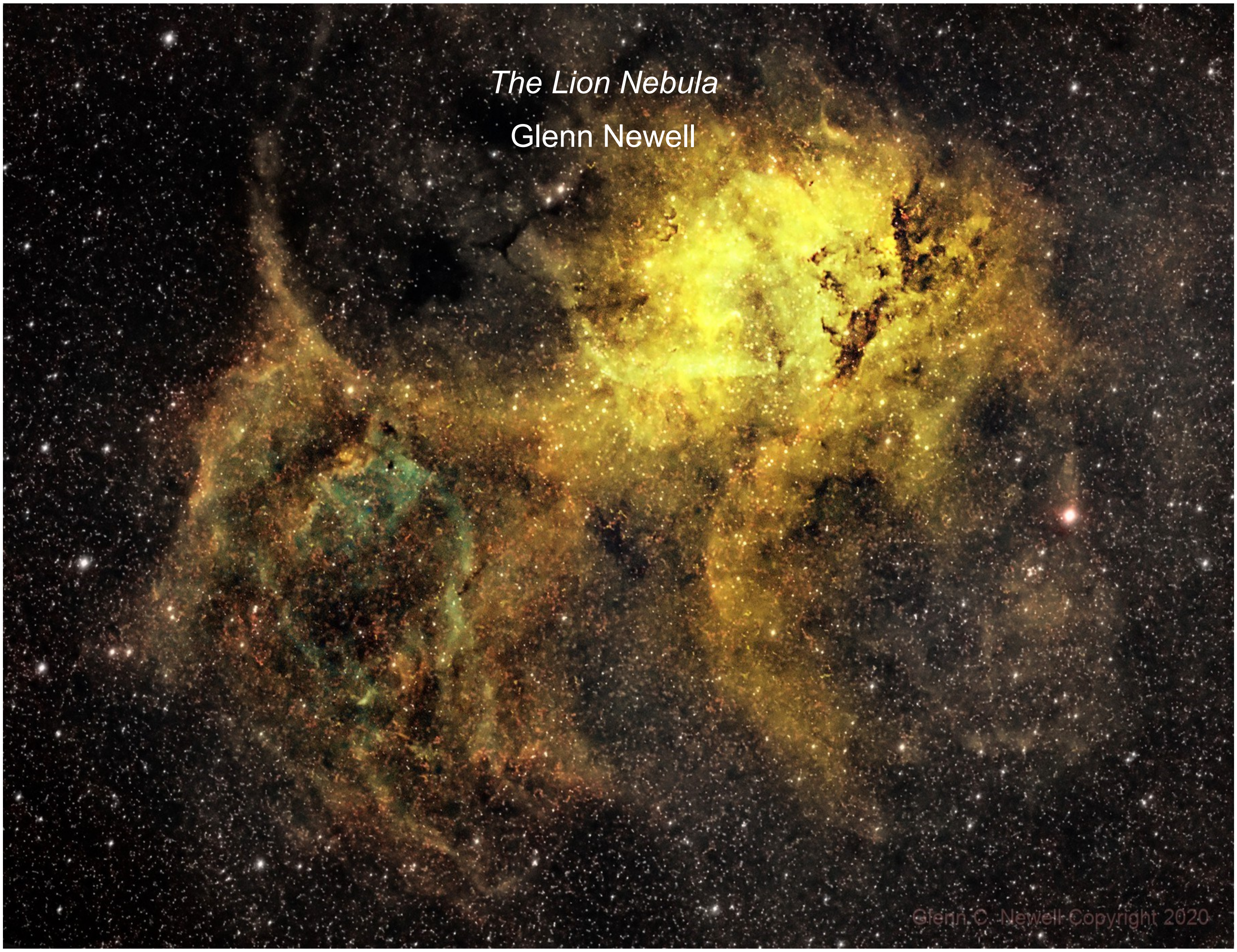
Telescope: Orion 10" f/3.9 Astrograph

Camera: QSI 683 wsg-8

Mount: Software Bisque Paramount MYT

Software: Software Bisque Sky X Pro , PixInsight 1.8 PixInsight , Seqence Generator Pro , PHD2 Guiding

SJAA Member Astrophoto Gallery



Dates: Sept 7, 2020

Location: Union City CA

Telescope: William Optics Megrez 80 ED II Triplet APO

Camera: ZWO ASI1600MM-Cool

Mount: Orion HDX110 EQ-G

Software: PixInsight PI 1.8, Adobe PS CC

SJAA Member Astrophoto Gallery

Date: August 14, 2020

Location: Henry Coe Park, San Jose, CA

Telescope: None

Camera: Nikon D750 with Kit lens 24-120 at F4, 15s, 3200 ISO. About 9 shots stacked and no tracker

Mount: None

Software: Sequator, Luminar 4 trial version and dark table

SJAA Member Astrophoto Gallery

Date: June 17, 2020; June 18, 2020; June20-22 2020

Location: Dark Sky New Mexico, Animas, New Mexico

Telescope: PLANEWAVE 14-INCH CDK

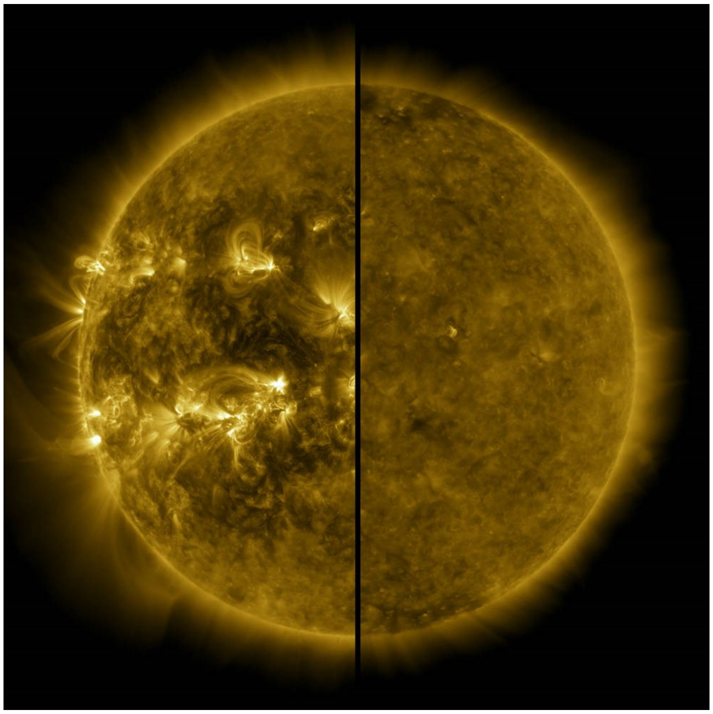
Camera: Finger Lakes Instrumentation ML50100

Mount: 10 Micron GM 3000 HPS

Software: Sequence Generator Pro, PixInsight1.8

Solar Cycle 25 Is Here. NASA, NOAA Scientists Explain What That Means

Credit: NASA



This split image shows the difference between an active Sun during solar maximum (on the left, captured in April 2014) and a quiet Sun during solar minimum (on the right, captured in December 2019). December 2019 marks the beginning of Solar Cycle 25, and the Sun's activity will once again ramp up until solar maximum, predicted for 2025.

Credits: NASA/SDO

Solar Cycle 25 has begun. During a media event on Tuesday, experts from NASA and the National Oceanic and Atmospheric Administration (NOAA) discussed their analysis and predictions about the new solar cycle – and how the coming upswing in space weather will impact our lives and technology on Earth, as well as astronauts in space.

The Solar Cycle 25 Prediction Panel, an international group of experts co-sponsored by NASA and NOAA, announced that solar minimum occurred in December 2019, marking the start of a new solar cycle. Because our Sun is so variable, it can take months after the fact to declare this event. Scientists use sunspots to track solar cycle progress; the dark blotches on the Sun are associated with solar activity, often as the origins for giant explosions – such as solar flares or coronal mass ejections – which can spew light, energy, and solar material into space.

"As we emerge from solar minimum and approach Cycle 25's maximum, it is important to remember solar activity never stops; it changes form as the pendulum swings," said Lika Guhathakurta, solar scientist at the Heliophysics Division at NASA Headquarters in Washington.

NASA and NOAA, along with the Federal Emergency Management Agency and other federal agencies and departments, work together on the National Space Weather Strategy and Action Plan to enhance space weather preparedness and protect the nation from space weather hazards. NOAA provides space weather predictions and satellites to monitor space weather in real time; NASA is the nation's research arm, helping improve our understanding of near-Earth space, and ultimately, forecasting models.

Space weather predictions are also critical for supporting Artemis program spacecraft and astronauts. Surveying this space environment is the first step to understanding and mitigating astronaut exposure to space radiation. The first two science investigations to be conducted from the Gateway will study space weather and monitor the radiation environment in lunar orbit. Scientists are working on predictive models so they can one day forecast space weather much like meteorologists forecast weather on Earth.

"There is no bad weather, just bad preparation," said Jake Bleacher, chief scientist for NASA's Human Exploration and Operations Mission Directorate at the agency's Headquarters. "Space weather is what it is – our job is to prepare."

Understanding the cycles of the Sun is one part of that preparation. To determine the start of a new solar cycle, the prediction panel consulted monthly data on sunspots from the World Data Center for the Sunspot Index and Long-term Solar Observations, located at the Royal Observatory of Belgium in Brussels, which tracks sunspots and pinpoints the solar cycle's highs and lows.

"We keep a detailed record of the few tiny sunspots that mark the onset and rise of the new cycle," said Frédéric Clette, the center's director and one of the prediction panelists. "These are the diminutive heralds of future giant solar fireworks. It is only by tracking the general trend over many months that we can determine the tipping point between two cycles."

With solar minimum behind us, scientists expect the Sun's activity to ramp up toward the next predicted maximum in July 2025. Doug Biesecker, panel co-chair and solar physicist at NOAA's Space Weather Prediction Center (SWPC) in Boulder, Colorado, said Solar Cycle 25 is anticipated to be as strong as the last solar cycle, which was a below-average cycle, but not without risk.

Kid Spot

Kid Spot Jokes:

Q: Why does the moon orbit the Earth?

A: To get to the other side??

Q: How do astronomers organize a party?

A: They planet.

Q: What music do Astronauts listen to?

A: Nep-Tunes!

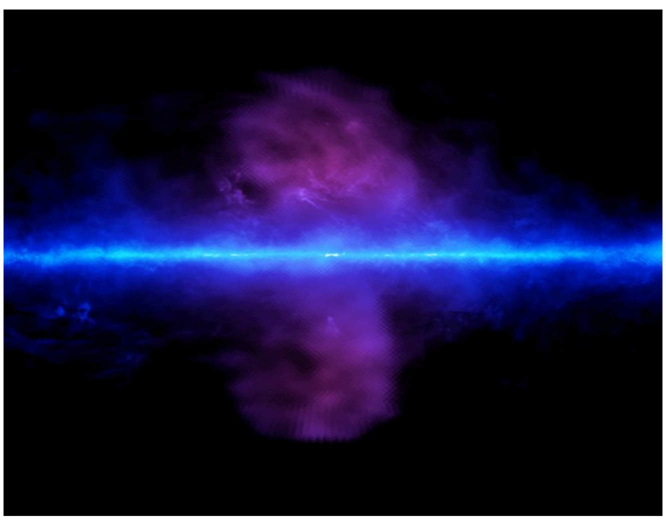
Anything that doesn't matter has no mass.

How many general relativists does it take to change a light bulb? Two. One holds the bulb, while the other rotates the universe.

Why did Werner Heisenberg hate driving cars? Because, every time he looked at the speedometer he got lost!

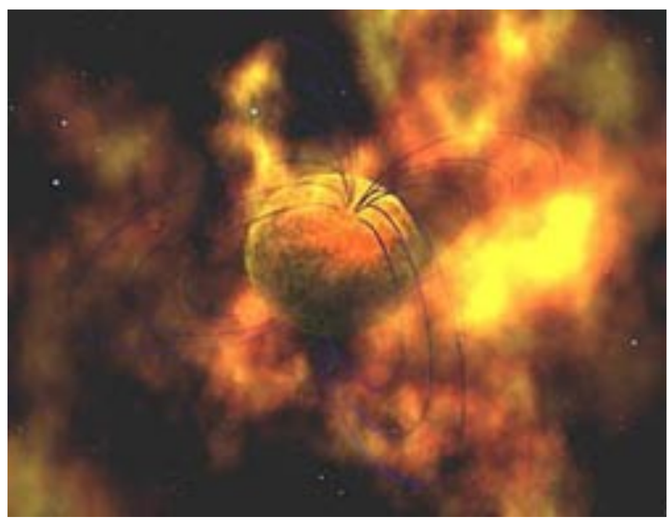
Did you Know?

Fermi Bubbles



- The Fermi bubbles are two large structures in gamma-rays above and below the Galactic center
- The plane of our galaxy (shown in blue above) glows brightly in gamma rays, which result when high-energy particles called cosmic rays interact with gas and dust.
- The bubbles may be related to the release of vast amounts of energy emitted from the supermassive black hole at the center of our Milky Way galaxy.

Magnetar



- Ultra-powerful magnetic neutron stars play hide-and-seek with astronomers. They're known to erupt without warning, some for hours and others for months, before dimming and disappearing again
- Magnetars, the most magnetic stars known, aren't powered by a conventional mechanism such as nuclear fusion or rotation.
- Magnetars have magnetic fields a thousand times stronger than ordinary neutron stars that measure a million billion Gauss, or about a hundred-trillion refrigerator magnets!

From the Board of Directors

Announcements

- ◊ San Jose Astronomical Association is closely monitoring the coronavirus/COVID-19 situation in Santa Clara County. In accordance with guidance from Santa Clara County Health Department, SJAA leaders have been canceling many events, especially those that involve close contact and shared surfaces on telescopes. Some events, however, have successfully been moved to an online format. Please check our Meetup to determine if an event has been canceled or moved online. The board of directors will periodically review the status of the outbreak and the guidance from the county to determine the impact upon upcoming events.
- ◊ New board member elections will be held at the February 2021 board meeting on February 27. The following 4 board positions will be open for Elections: Director 1, Director 3, Director 5, Director 7, and Director 9. All current incumbents are expected to run for their positions again. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.
- ◊ New officer elections for President, Vice-President, Treasurer and Secretary positions will be held at the March 2021 board meeting on March 27. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.
- ◊ 2021 Calendar is ready
- ◊ The Fall Swap Meet is canceled and the next will be the Spring Swap Meet

Board Meeting Excerpts

October 3, 2020

In attendance

Swami , Rob J, Glenn, Wolf, Vini, Peter, Sukhada, Emi
Excused Absences: Ken, Sandy
Guests: Kanch, Marianne

2021 Events Calendar

Dave Ittner and the board are working on the 2021 club calendar. It is almost done.

Projects for Volunteers

SJAA still needs board members and program leaders to provide ideas for projects that may be done by volunteer members.

Houge Park Cleanup

The city of San Jose, asked SJAA to prepare building 1 for the upcoming national elections. Kanch, Akshay, Glenn and Nikola went on October 10 to clean up and organize building 1. For additional information and photo please look at page 2.

YouTube Copyright Claims

SJAA has gotten tagged for copyright claims on YouTube videos. Videos got flagged for audio copyright infringement

November 7, 2020

In attendance

Swami, Sukhada, Wolf, Vini, Rob J, Glenn Ken, Emi, Sandy, Peter
Excused Absences: none
Guest: Marianne, Kanch

2021 Calendar Review

During the pandemic time ITSP and Starry Nights are on hold and are to be replaced with the Armchair Star Party. All the meetup events are displayed on the SJAA website on the right side.

Loaner Program Resurrection Plan

President Ken Miura was to send a letter to the city of San Jose for requesting access to Houge Park building 1.

Armchair Star Party November Event

Last armchair star party for the year was held on November 21st. A few interesting objects were selected for the showcase to be previewed such as Mars. A slide of the presenters during that show is included on page 2 of this issue.

PST Solar Loaner Scope

SJAA received a PST Solar Telescope from Marilyn Perry as a donation.

December 5, 2020

In attendance

Ken, Swami, Wolf, Vini, Sukhada, Glenn, Sandy, Peter, Emi
Excused Absences: Rob J
Guests: Kanch, Marianne

Online voting for board and officer elections

Due to COVID-19 constraints the February and March 2021 board and officer elections are going to be held online. Additional information will be announced later.

Open Vice President Position

The position of Vice President is open. The remaining term for this position is until March 27, 2021.

Armstrong Scholarship Donation

Heidi Gerster and her late husband Isaac Kikawada donated $500 towards the Armstrong fund.

Fall Swap Meet

Fall swap meet was canceled due to COVID-19. Next, swap meet is the Spring which is tentatively scheduled for April 25 subject to the Covid-19 situation and restrictions in place at that time.

Loaner Program Resurrection Plan

SJAA has decided that it is not a good time to make a request to the city for access to Houge Park due to COVID-19 and the current status of Santa Clara County.

Ephemeris Color Printing Questionnaire

On page 15 of this issue there is an Ephemeris color printing questionnaire. Based on the feedback from the readers SJAA will decide should the Ephemeris be in either full colored pages, or 4 pages of astroimages color printed or fully black and white color printed. This form is open until end of February 2021.

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

http://www.meetup.com/SJ-Astronomy/

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is October 12th.

http://www.sjaa.net/programs/quick-start/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.
If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.
Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill. Contact SJAA's (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

President/Dir:	Ken Miura
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary:	Emi Nikolov
Director:	Sandy Mohan
Director:	Peter Melhus
Director:	Vini Carter
Director:	Swami Nigam
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Abishek
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal
Publicity:	Rob Jaworski
Quick STARt	Dave Ittner
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Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Manager:	Beth
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

E-mails: http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

Ephemeris Printing Questionnaire

Please help us the Ephemeris production team by providing your input to the following questions. You may do so by either going to this URL: https://tinyurl.com/yxaqjhmq or by filling it in and mailing it to: SJAA, P.O. Box 28243 San Jose, CA 95159-8243.

This form is open until the end of February 2021.

1. Your Name (Optional):

2. In the last few editions, we moved the Astronomy pictures to the center 4 pages and made them colored. What do you think of this change:

a) Love it

b) Hate it

c) Didn't care or I don't get a printed copy

d) Didn't even notice the change

3. We have explored the cost of printing Ephemeris in 3 possible formats. Color printing significantly increases the cost, so it will require us to increase the annual amount we charge for printed newsletter. Note, the full color PDF version is always available on our website. Please indicate which of the following 3 options do you prefer:

a) Full B&W printing - member cost stays at $10 per year

b) Print 4 pages with astroimages in color, other pages B&W - member cost increases from $10 to $15 per year

c) Print all pages in color - member cost increases from $10 to $20 per year

4. If there are any other suggestions regarding Ephemeris that you would like to share with us, please write them here:

Place
postage
here

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P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: http://www.sjaa.net/sjaa-newsletter-ephemeris/

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: http://www.sjaa.net/join-the-sjaa/

Name: ______________________________

Address: ______________________________

City/ST/Zip: ______________________________

Phone: ______________________________

E-mail address: ______________________________